Rule-Based Pattern Matching for FAQ Chatbot Systems: A Keyword Matching Approach for University Student Support

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Abstract. This report explains a rule-based FAQ chatbot for university student support using keyword pattern matching. A scoring system calculates how well queries match, handle partial matches, and use priority-based tie-breaking. Testing show it works 86.7% of the time when the confidence threshold is optimally configured. The report highlights the benefits of rule-based methods, like being easy to understand, but also mentions challenges such as struggling with changes in meaning. Comparing it to Naive Bayes and TF-IDF methods shows the pros and cons of different chatbot designs.

1 Introduction

Chatbots have become prevalent in educational institutions for automating student support queries. This report implements a rule-based FAQ chatbot using keyword pattern matching to assist university students with COMP1827 coursework questions about deadlines, submissions, and marking criteria. Rule-based systems use predefined patterns and explicit if-then logic, offering transparency and ease of debugging compared to machine learning approaches, making them suitable for domain-specific applications with well-defined query types.

2 Background

Rule-based chatbots originated with ELIZA (Weizenbaum, 1966), using pattern matching to simulate conversation. The architecture consists of a knowledge base storing question-answer pairs, a matching engine comparing user input against patterns, and a response generator. Keyword-based matching identifies trigger words regardless of context, while confidence scoring handles multiple rule matches by assigning numerical values based on matched keywords and predefined priorities. Advantages include transparency and no training requirements, though limitations include manual rule creation effort and inability to handle out-of-pattern queries.

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3 Experiments and Results

3.1 System Architecture

The implemented chatbot has four key parts: a knowledge base, a preprocessing module, a scoring engine, and a response generator. The knowledge base contains 18 FAQ entries covering topics including deadlines, submission requirements, marking criteria, plagiarism policies, and contact information. Each entry is structured as follows:

```
"deadlines": {
    "keywords": ["deadline", "due date",
                 "when due", "submission date
                    "],
    "response": "The presentation is on...",
    "category": "assessment",
    "priority": 3
}
```

This structure allows multiple keywords to trigger the same response, categorisation for analytical purposes, and priority values for tie-breaking when multiple entries achieve similar match scores.

3.2 Matching Algorithm

The matching process begins with preprocessing user input through lowercase conversion and punctuation removal to ensure consistent comparison. The scoring algorithm then evaluates each knowledge base entry against the processed input using the following approach:

$$S_{total} = \sum_{k \in K} \begin{cases} 1.0 + B_{exact} + L_k & \text{if exact match} \\ 0.5 \times \frac{W_{matched}}{W_{total}} & \text{if partial match} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Where K represents the keyword set for an entry, B_{exact} is the exact match bonus (0.5), L_k is a length factor rewarding longer phrase matches, $W_{matched}$ is the count of matched words in multi-word keywords, and W_{total} is the total words in the keyword. The final score is normalised by dividing by the number of keywords:

$$S_{final} = \min\left(\frac{S_{total}}{|K|}, 1.0\right) \quad (2)$$

3.3 Parameter Configuration

The system includes four tunable parameters that affect matching behaviour:

- **CONFIDENCE_THRESHOLD** (0.3): Minimum score required to accept a match. Values below this threshold trigger the fallback response.
- **EXACT_MATCH_BONUS** (0.5): Additional score awarded for exact keyword matches.
- **ALLOW_PARTIAL_MATCHING** (True): Enables matching individual words within multi-word keywords.
- **MIN_WORD_LENGTH** (3): Filters short words that provide little discriminative value.

Table 1 shows results across different threshold values.

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Threshold	Correct	Incorrect	No Response
0.1	14	0	1
0.3	13	0	2
0.5	6	0	9
0.7	2	0	13
0.9	1	0	14

Table 1. Effect of confidence threshold on response accuracy

Although threshold 0.1 achieved 14/15 correct (93.3%), threshold 0.3 was selected as optimal with 13/15 correct (86.7%) to balance coverage and precision. Lower thresholds risk false positives with ambiguous queries, while higher thresholds (0.5, 0.7) become overly conservative, rejecting valid questions. This demonstrates the fundamental trade-off in rule-based systems between answering more questions and avoiding incorrect responses.

3.4 Performance Analysis

The logging system keeps track of every interaction and helps calculate performance metrics. When tested, it showed steady behavior with average confidence scores of 0.72 in identifying successful matches. Students most often triggered intents like deadlines and marking criteria, which reflect their usual concerns.

Looking at failed queries showed some issues with dealing with synonyms and rephrased questions. For instance, the phrase What's the cutoff date?" did not match the deadlines category because the word "cutoff" was missing from the list of keywords. This shows how rule-based systems rely on predicting the words users might use, which adds to the upkeep required.

3.5 Comparison

There are clear trade-offs when comparing this rule-based strategy with alternative approaches. Through probabilistic learning, the Naive Bayes classifier provides better generalization to unseen phrasings; however, it requires labeled training data and may yield unexpected classifications. Although it lacks the interpretability of explicit rules, the TF-IDF approach uses vector similarity to handle vocabulary variations more gracefully. Although it requires a lot more processing power and training data, the neural network exhibits contemporary deep learning concepts and is capable of learning intricate patterns. The primary benefit of rule-based pattern matching over machine learning techniques is total transparency the matched keywords and confidence scores unmistakably show decision reasoning—despite the fact that it achieved 86.7% accuracy. It cannot, however, adjust to new query patterns without manual keyword updates, in contrast to learning-based approaches.

3.6 Legal, Social, Ethical and Professional Issues

Using chatbots to assist students brings up various LSEPI concerns. the system needs to manage personal data under GDPR rules when logging can identify specific users. Social concerns include the risk that relying too much on automated systems might limit human interactions and affect students who find text-based communication difficult. There are ethical considerations as well, like making sure chatbots are recognized as automated tools and that they pass on issues to humans when they cannot handle them. On a professional level, developers have a responsibility to deliver accurate information since wrong advice about deadlines or requirements could harm students' success.

3.7 Limitations

The biggest weakness of rule-based methods is their struggle with any changes in language. These systems only pick out set patterns and do not grasp meaning. They miss the mark when there are typos new words, or unusual ways of saying things even if what the user wants is obvious. It also takes constant manual work to keep keyword lists up-to-date as new ways of asking questions appear. Since the system cannot learn on its own, it stays the same unless someone updates the rules.

4 Conclusion and Future Work

This report explained a rule-based FAQ chatbot built to help university students. It used keyword pattern matching paired with confidence scoring and selected responses based on priority. The chatbot answered questions about coursework deadlines, submission details, and marking rules, achieving 86.7% accuracy

The project showed the benefits and drawbacks of using rule-based methods. The system is predictable and easy to understand but relies on adding keywords and struggles to handle anything outside its set patterns. Comparing this to machine learning models developed by team members showed a balance between making the chatbot easy to interpret and allowing it to adapt more .

Future updates may improve the system in various ways. Using stemming or lemmatisation can make it better at matching words with different forms. Expanding synonyms with WordNet might lower the effort needed to maintain keywords. Letting users rate responses could help spot missing information in the knowledge base. combining rule-based matching to tackle straightforward queries with machine learning for uncertain ones could take advantage of both methods .

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